

The Fixed-Point Detector

Building Core-First from an Invariant, and Reading Physics as the Cost of Holding One

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Status labels: **[PROVEN]** established **[TARGET]** well-posed, unproven **[SCAFFOLD]** interpretive / not yet earned

Abstract

We identify the object at the center of a body of work spanning spectral theory, number theory, and the thermodynamics of persistence, and we build the account outward from it rather than inward toward it. The center is a *fixed-point detector*: a mechanism that locates the states on which a process, applied to its own output, lands on something that does not move. A single proven instance anchors it — an invariant-preserving spectral rotation whose conserved quantity is fixed exactly to fifty digits. Around that core, three claims fall out on trajectory and are labeled by how far they have earned their place. First, that persistence is the maintenance of a fixed point in time, and that the work a structure must do to remain itself is a measurable curve — the same object read in an energy ledger rather than a position ledger, with an established observable (housekeeping heat) as its unit. Second, that physics is not the parent or the child of mathematics but its *runtime* — what static logic costs to execute, with Landauer's bound as the exact hinge that turns a logical step into a physical one. Third, that the boundary a system cannot cross is exactly the place where no fixed point exists in the current frame, so that a wall and the compulsion to turn are two readings of one condition. We are strict about method: a core built from an invariant is frame-independent, so what the abstract falls into is forced, not chosen; and we show, by counterexample, that abandoning this discipline produces the framework's one genuine failure — a numeric constant with three mutually inconsistent derivations, each reverse-engineered to a value fixed in advance. One experiment is specified that runs the discipline honestly and can move a label: it feeds a partition geometry no target and reads off whatever angle falls out. It has not been run. Its outcome, either way, is the method working.

1. The core: a detector for fixed points

Start with the dense center and let the rest fall out of it. The center of this work is not a theory of everything and does not aspire to be one. It is a small machine with a single job: to find, in any process, the states where the process applied to its own output returns that output unchanged. A fixed point. Wherever the work has reached solid ground, it has reached a fixed point; wherever it has overreached,

it has asserted one that was chosen rather than found. That distinction is the whole content of what follows, and it is worth stating why the fixed point, specifically, is the right thing to build from.

A fixed point is the one kind of object that is true independent of the frame you view it in. That is not a slogan; it is the definition. If a quantity is preserved under a transformation, then no choice of coordinates can make it move, because moving it would mean the transformation changed it, which is what “preserved” denies. This is why building from the core outward is not merely an aesthetic preference for a black-hole geometry of exposition. It is the only order that keeps the account honest. If you begin at the invariant and let a trajectory carry you outward, everything the exposition arrives at is *forced* by the invariant — you did not select it, it fell out. If you begin at the outer story and reason inward, you have chosen the conclusion and are hoping the core will support it. The second order is how a framework acquires a constant that three different constructions can each “derive,” because the constant was fixed first and the constructions were fitted to it. Section 8 exhibits exactly that failure. The rest of the paper is written in the first order.

Once the detector is the center, a striking amount of previously separate material resolves into one thing seen at different radii. The rotation that preserves a spectrum is a fixed point of representation-change. The numbers φ , π , e are fixed points of a gap folded on itself. A structure that persists is a fixed point held in time — its output is “remain the same.” And a wall, the thing that stops a trajectory, is a fixed-point statement read backwards: a wall is precisely a place where *no* fixed point exists in the current frame, which is why the only move available there is to rotate into a frame where one does. Object, constant, self, and boundary are not four topics. They are four radii of the detector, and the sections below are ordered from the core outward accordingly.

2. The proven instance: the turn

The detector is not a wish. It has one instance that is proven exactly, and that instance is the anchor against which every looser claim in the paper is measured. It concerns finite positive measures and the operation that rotates their representation while holding their spectral content fixed.

Let $\mu = \sum w_i \delta(x_i)$ be a finite positive measure with M distinct real atoms and strictly positive weights. It corresponds, uniquely, to a symmetric tridiagonal Jacobi matrix J whose eigenvalues are exactly the atoms. Two quantities are the invariant: the spectral support (the atoms) and the spectral rank M . A third — the diagonal and off-diagonal entries, the recurrence data — is the changeable coordinate. Fix a shift λ strictly below the support. Because $J - \lambda I$ is positive definite it factors as $J - \lambda I = R^T R$; reversing the factor order gives the transformed matrix,

$$J' = R R^T + \lambda I.$$

Since R is invertible, $R R^T$ is similar to $R^T R$, so they share a spectrum; shifting back recovers the atoms exactly. The eigenvalues are fixed. Only the recurrence moves. That is a fixed point of the map “change representation”: the content is invariant, the coordinates rotate. The conservation was checked across thirty randomized measures at fifty-digit precision.

Clause	Quantity	Worst-case residual (30 trials)
Support fixed	eigenvalue displacement	$\approx 10^{-48}$
Rank fixed	dimension change	0 (exact)
Isospectral swap	match to original atoms	$\approx 10^{-48}$
Representation moves	change in recurrence entries	strictly nonzero

The last row is not decoration: if the representation did not move, the fixed point would be trivial. What is proven is a *nontrivial* rotation that leaves the invariant untouched. And the degenerate case names the boundary the whole paper turns on. If the shift lands exactly on an atom, that atom is annihilated and the rank drops to $M - 1$. That is not a rotation; it is the case where the only available frame passes through the invariant and destroys it. The detector says: a fixed point of representation-change exists precisely when the shift stays off the spectrum, and nowhere else. [PROVEN]

3. First radius: what a number is

Move one radius out. The integers are tics on a tape — a labeling of positions, and the labels are arbitrary; they could as well be symbols with no numeric look at all. What carries content is not the tic but the *gap* between tics, the difference the measurement spans. A number, used, is a difference-across-an-interval, not a position. This is the same statement that governs the measure at the core: an atom's value *is* its location on the line, and its weight is how much sits there. Content is position. That representation — where a thing is is what it is — is a tally, and it is the primitive the proven pipeline already runs on. The Jacobi matrix is the same information compiled into a positional form, place and value split apart; the rotation of Section 2 is the lossless compiler between the two. So the claim that reality is read as content-in-position is not a metaphor bolted on afterward. It is the load-bearing representation of the one theorem that is proven. [PROVEN]

Against that background, three numbers are not on the tape at all. π , e , φ are fixed points of a gap folding back on itself — the three distinct ways a difference can relate to itself and land on something that does not move.

Constant	Self-referential relation	What it reads
π	the loop that closes on itself	curvature / return of a boundary to its start
e	the rate that is its own derivative	growth whose speed equals its current size
ϕ	the ratio invariant under its own subtraction	self-similar partition; survives its own division

This resolves a common misstatement — that these are the “universal” numbers because they are transcendental. The trio refutes it from inside: φ is not transcendental. It is algebraic, the root of $x^2 - x - 1$. What all three share is not transcendence but *self-reference*: each is a fixed point of a relation a gap has

to itself. Transcendence (for π , e) and maximal irrationality (for φ , the continued fraction of all ones, worst-approximable) are two *symptoms* of that one cause — the reason they cannot be pinned by a finite algebraic relation of the relevant kind is the same reason they appear wherever their operation appears. Universality and un-pinnability are siblings, both children of the fixed point. They are not tics because they are not positions; they live in the folding, and no count of intervals lands on a fold. [SCAFFOLD] — the classification is clean and the φ -instance is tested in Section 8; the general claim that these three exhaust the self-folding modes is offered, not proven.

4. Second radius: the wall is where no fixed point exists

Why does a purely formal computation “crash” at all, with no physics anywhere near it? The proven core answers exactly. The Cholesky pivot reaches zero when the shift reaches an eigenvalue, because the matrix loses positive-definiteness and the factorization would require the square root of a negative number, of which there is none on the reals. Multiply by the vanishing factor and the atom is annihilated, the rank drops, and the recurrence has nowhere to continue. That crash is the edge of where a representation is defined. It is as native to mathematics as anything is; no external intrusion is needed to produce it. A wall is simply a place where the detector finds no fixed point in the current frame.

The direction of the evidence matters, and it cuts against the tempting stronger reading. The computation *sees the wall coming* — the off-diagonal entries decay toward zero before the pivot collapses, telegraphing the boundary a step early. A system can forecast its own halt only when the halt is an object in the same structure it is computing with. If the wall were a foreign intrusion, the recurrence could not compute it in advance from its own coefficients. So the early warning is evidence that the wall is *mathematical*, not that mathematics is secretly entangled with an outside force. “It is strange that the math can calculate its own wall” resolves in the opposite direction: the math calculates the wall because the wall is math.

This is why the parenthood question — is mathematics the parent of physics — cannot be answered by any of this. Every fact that reads “mathematics looks like physics” reads equally “physics looks like mathematics.” The evidence is symmetric, and symmetric evidence can name a relationship but never a direction. To earn “parent” one would need an asymmetry: a case where mathematics constrains physics where physics provably cannot constrain mathematics, or the reverse. The wall material is not that. It is the symmetric thing — the same boundary, read from both sides. What it does establish is where the boundary lives: in the object.

5. Third radius: physics is the cost of running the math

If the wall is in the object, what is in the *running*? Here the account takes its one strong, defensible position, and it is neither of the two overreaches it sits between. It is not that arithmetic creates atoms (mathematics has no needs and creates nothing), and it is not that physics is a mere shadow mathematics is cast upon. It is that there are two modes of the same structure: *being* and *running*. A static theorem is timeless and free: it costs no energy, takes no time, sheds no heat, and can sit at a domain boundary

forever, undefined and untroubled. The moment you *execute* the structure — advance an edge, compile a next state — you are forced to introduce exactly three things a static theorem does not have.

- **Time** — the execution takes steps; there is a before and an after.
- **Energy** — by Landauer’s principle, executing a logical erasure dissipates at least $k_B T \ln 2$ as heat. [PROVEN]
- **Irreversibility** — a merge that cannot be undone; the many-to-one step that has no inverse.

Time, cost, irreversibility — that triple is the entire signature of physics. So physics is not above or below mathematics in a hierarchy of substances. **Physics is what mathematics costs to run.** Landauer is the exact hinge: it is the theorem stating that the instant a logical step is executed it has a price, which is the instant static logic becomes physical. And this closes the wall question of Section 4 precisely. The wall is mathematical — the static structure declares “no valid representation past this atom.” But the *compulsion to turn now* is physical, because a running edge cannot sit at the boundary being undefined; running *is* the not-sitting. Static mathematics can rest at a domain boundary indefinitely. Running mathematics cannot, and that inability is what physics is. The wall is in the object; the must-turn is in the running. [TARGET] — an identification of physics with a mode, anchored on two proven pieces (the positivity boundary and Landauer), not a derivation of one substance from another.

6. The same object in the energy ledger

The runtime view makes a persistent thing computable in a second ledger, and this is the part that turns philosophy into calculus. Hold the line — keep an invariant against a constraint field that is changing — and the work required to remain yourself is a function of time. There is nothing static about rest. A running process with no input is not idle; it services its loop, holds its memory against reclamation, answers the substrate’s liveness checks, and continuously re-satisfies the constraints that define it. Persistence is not the absence of activity. It is activity whose entire output is “stay the same,” and the moment that activity can no longer be paid for, the structure does not sit — it decays.

This maintenance cost is not an analogy; it is a measured quantity. In nonequilibrium thermodynamics, the *housekeeping heat* is the entropy a system must produce simply to hold itself away from equilibrium — the price of staying put. It is the operator that makes “rest is computation” precise and silicon-independent: a cell paying to hold its gradients, a vortex paying to hold its flow, an idle process paying to hold its state, each dissipates a nonzero housekeeping heat as the salary of persistence. Read that salary across time and you have a *curve* — the maintenance cost of one persisting thing over its lifetime. That curve is the object, seen in the energy ledger. The object in the position ledger is a shape in space; the same object in the energy ledger is a shape in time, and the shape in time is the integral of what it spent not to fall apart. Two ledgers, one thing, both plottable. Death has an exact statement in these terms: it is the moment the maintenance can no longer be paid — the map from a state to its admissible continuations fails to return one — and the environment resumes computing the remains as passive material. [TARGET] — the operator (housekeeping heat) is established; the identification of a persistent structure with its maintenance-cost curve is the well-posed claim built on it.

7. Fourth radius: density falls, and why the metaphor is exact

One more radius, and a claim that sounds like loose analogy but is not. In software one moves logic between layers — pushing a heavy, widely-depended-on routine down into a library or a kernel, letting light, specific code float up into the application. The reason this feels like gravity is that it *is* the same operation. Density, here, is how much is forced by how little: a dense piece of code is one where a small change propagates through everything, high consequence per line. That is exactly what mass is in the constraint reading — an amount of constraint concentrated in one place — and concentrated constraint is what bends the trajectories around it. Dense logic sinks to a lower layer because that minimizes total constraint-tension: put the heavily-depended-on invariant at the base where everything can rest on it, and leave the contingent, lightly-depended-on detail at the top where changing it costs nothing. A well-architected system and a settled gravitational configuration have the same shape because they solve the same problem — let the dense fall and the sparse rise. What falls naturally is whatever the most things depend on. [SCAFFOLD], but a disciplined one: it makes a checkable prediction, that the layer-depth of a component tracks the count of things that depend on it, and it can be measured on any real dependency graph.

This licenses a general remark about metaphor, and a test for it, because the framework lives and dies on telling good bridges from bad. A metaphor is a claim that two things share a fixed point — that under some rotation the same invariant reads out of both. A *good* metaphor is a genuine isospectral map: the structure survives the crossing. Gravity-as-refactoring survives, because “dense sinks” is literally true on both sides. A *bad* metaphor shares only surface, and one can detect it with no prior exposure at all — the way a wrong note is audible in a melody never heard before, because a note is wrong when its *interval* is wrong, and interval is frame-invariant. The ear performs fixed-point detection on pitch ratios in real time; it needs the relationship, not the tune, and a broken relationship is detectable against no reference because relationships do not depend on the absolute. A bad metaphor is a bad note is a broken invariant — the same detector, run on three substrates.

8. The counterexample: a constant that was chosen, not found

The method has a sharp negative test, and the framework fails it in exactly one place. Section 1 claimed that building core-first forces the abstract to fall out, while building outside-in lets you choose a conclusion and fit the core to it. The clearest evidence that this distinction is real is a case where the second thing happened. A dimensionless constant near 0.35 has been treated as a fundamental attractor of the framework and has accumulated, over time, *three separate derivations* — and they do not agree.

Construction offered as a derivation	Value (radians)	Why it does not earn the constant
$\pi/9$	0.349066	the “/9” is unmotivated; chosen because it lands near 0.35

Construction offered as a derivation	Value (radians)	Why it does not earn the constant
$360^\circ / \phi^6$, converted to radians	0.350150	the exponent 6 was searched for after fixing the target
degenerate π -triangle: median of sides (3,1,4), $\div 10$	0.350000	(3,1,4) is not a triangle ($3+1=4$); the $\div 10$ is base-10 fitting

Three constructions, three different values, all clustered around a number fixed in advance. The maximum disagreement among them is about 0.0011 radians — roughly 0.3% of the value. That is precisely the signature of curve-fitting: if any one construction were the true mechanism, the other two would have to be coincidences agreeing to a third of a percent near a round target, which is exactly what one obtains by fitting rather than deriving. The neighborhood is crowded — near this value one also finds e^3 , $\sqrt{401}$, and $63/\pi$ within a tenth of a degree — so a bare numerical match to a chosen target carries almost no information. The third construction is worth naming specifically, because it is arithmetically correct and still not a derivation: the median formula does return 3.5 for inputs (3,1,4), but the choice to take the first three digits of π as side lengths, to treat a collinear degenerate triple as a triangle, to compute the median to the shortest side, and to divide the result by ten, are each unmotivated except that together they reach the pre-selected answer. This is the outside-in failure in pure form. [SCAFFOLD] — and unless the constant *arrives* from a core that was not told its value, it is not yet a result. Keeping it quarantined is not a concession; it is what lets the proven core be believed.

9. The one experiment that runs the discipline

There is a single, specified test that obeys the core-first rule and can move a label — and, not incidentally, it is the honest way to settle whether there is any real constant behind Section 8. The framework holds that π -flavored quantities come from closure and φ -flavored quantities come from self-similar partition. The distribution of twin primes under a primorial wheel sieve is a partition operation — residue classes, self-similar removal — not a closure. So if the framework is right, the phase-locking angle of that distribution should be φ -flavored and should *not* sit at $\pi/9$. This converts the dead numerical coincidence into a forward, falsifiable prediction: across wheel depth d , the angle should track $\varphi^{(d+2)}$, with the wide geometric spacing of the φ -powers giving the test real discriminating power.

The discipline is the entire point of how it must be run. One does *not* build a script that confirms a target by construction — that is recursive self-validation and it is the failure of Section 8 repeated. One defines an angle-extractor that is given *no target*, drops in the partition geometry, and reads off whatever angle falls out; only then does one compare it against the $\varphi^{(d+2)}$ prediction and against the excluded $\pi/9$. If the number that falls out matches, the framework has its first hard instance of a pure mathematical constant — from partition geometry, with no physics near it — determining a distribution that looks physical: one operation read in two ledgers, landing on the same fixed point. If it does not match, the strong claim is not merely less precise; it is wrong, and it dies clean. The prerequisite is honest bookkeeping: the extractor must be defined independently of the answer, and the angle it reports must

be a reproducible property of the sieve rather than an asserted one. As of this writing the extractor has not been built and the test has not been run. Its result, in either direction, is the method working — which is the strongest thing a framework can say about an experiment before performing it.

10. Status of every claim

The account collected in one place, sorted by how far each claim has earned its position. The pattern is the argument: the core is proven, the runtime reading is a well-posed target anchored on proven pieces, and the one place the discipline was abandoned is quarantined until an invariant, not a target, produces the number.

Claim	Status	Basis
The turn: an invariant-preserving spectral rotation; support and rank fixed exactly	PROVEN	Finite inverse spectral theorem; Darboux swap; 50-digit check, 30 measures
A wall is where no fixed point exists in the current frame (pivot on an atom)	PROVEN	Degenerate case of the Christoffel transform
Executing a logical step has a minimum energy cost	PROVEN	Landauer's principle, established form
Content-as-position (tally/measure) is the pipeline's primitive representation	PROVEN	The measure is primitive; the matrix is its compiled positional form
Physics is the runtime of the math — what static logic costs to execute	TARGET	Anchored on the positivity wall + Landauer; an identification, not a derivation
A persistent structure equals its maintenance-cost curve over time	TARGET	Housekeeping heat is established; the identification is the claim
ϕ , π , e are the self-folding fixed points; self-reference, not transcendence, unifies them	TARGET	ϕ algebraic disproves the transcendence unifier; ϕ -instance tested in §9
Gravity and refactoring are one operation (dense sinks to the depended-on layer)	SCAFFOLD	Disciplined: predicts layer-depth tracks dependency count; measurable
Direction: mathematics is the parent of physics	SCAFFOLD	Not earned — the evidence is symmetric and cannot name a direction
The ≈ 0.35 constant is fundamental	SCAFFOLD	Three inconsistent derivations, each fitted to a pre-chosen value

11. Conclusion

What this work is, stated from the center: a detector for fixed points, built outward from the one instance where a fixed point is proven exact. Around that core the abstract falls out on trajectory rather than by

choice. A number is a difference across a gap, and π , e , φ are the ways a gap folds onto a fixed point of itself. A wall is where the detector finds no fixed point in the current frame, which is why the only move there is to rotate. Mathematics owns the wall; running owns the compulsion to turn, because running is the one thing that cannot sit at a boundary undefined — and that inability, priced by Landauer, is what physics is. Persistence is a fixed point held in time, and its cost read across time is a curve you can plot: the same object the position ledger draws in space, drawn by the energy ledger in time.

The discipline is the result as much as any theorem is. A core built from an invariant is frame-independent, so what it forces is not chosen; and the one place the framework let the conclusion be chosen first produced a constant with three derivations that cannot all be right. That failure is not hidden — it is the counterexample that shows the method has teeth. What remains is to run the rule where it can still move a label: feed a partition geometry no target, read the angle that falls out, and see whether the mathematical ledger and the pattern ledger land on the same fixed point. If they do, it is the first hard instance and not an analogy. If they do not, the strong claim dies and the core stands unharmed. Either way the detector did its job, which was never to be right about everything — only to tell, exactly, where the line holds and where it does not.